

Summary

Qingyuan YAO

Graduate from the AI master's program at Sorbonne University. Experienced in Unity with C# and AI algorithms with Python. Capable of implementing HCI and AI related theories into simulated environment.

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Portfolios	https://mrzny.github.io/
Github	https://github.com/MRVNY

Education

2021 - 2023	Master of Artificial Intelligence <i>Parcours AgeNts Distribués, Robotique, Recherche Opérationnelle, Interaction, DEcision (ANDROIDE)</i> Mention Bien Sorbonne University
2018 - 2021	Bachelor of Computer Science Mention Bien Sorbonne University

Experience

Deep Generative AI Internship

#Python #PyTorch #VAE #DeepLearning

Tokyo National Institute of Informatics

March - August 2023

- Created a new dataset of audio-text-motion by extracting motion data from dance videos using existing deep learning models of 3D pose estimation
- Implemented a Transformer VAE baseline model to showcase the validity of adding lyrics as a condition in the process of dance motion generation.

Serious Game Development Internship

#Unity #C# #SeriousGame

Team MOCAH of LIP6 at Sorbonne University

June - July 2022

- Solo developer of on the prototype targeted for PC and Android, developed by previous team of students
- Significantly improved stability and optimization by:
 - Fixing collisions and sorting layer issues
 - Using Coroutine, async for asynchronous execution
 - Using basic Jobs System and DOTs for multi-threading
 - Defining better structure with Heritage and Singleton
- Added new implementations into the game such as:
 - A visual novel interface with loadable conversation tree
 - Custom 2D A* pathfinding system
 - Character instant switching system
 - More extensive save&load system
- Communicated with the supervisor on a daily basis and proposed new implementation ideas

Game Design Workshop

2019 - 2022

- Learned the basics of game design
- Made 2 board-game prototypes (cf. portfolio)

Skills

- Programming:** Python, C, C#, C++, Java, JS, LaTeX
- Game development**
 - Unity (Experienced):** Targeted for desktop, mobile, web, structured C# code, level design, async, multi-threading, etc.
 - Unreal Engine 5 (Beginner):** Targeted for desktop, code with C++ and Blueprint
- Serious Games & Human-Computer Interaction:** Entity-Component System, Design Protocols, User Tracing, User Modeling, Adaptation, etc.
- AI:** Deep/Reinforcement Learning, Multi-Agent System, Game Theory, Operational Research, Decision
- Software and Web Development:** Object-Oriented Programming, Design Patterns, UML, basic frontend with Javascript, HTML, CSS
- Spoken languages:** Chinese (native speaker), English (C2), French (C1/C2), Spanish (A2/B1), Japanese (A2)
- Media Production:** Lightroom, Final Cut, Logic (limited competence: Procreate, Blender, Photoshop)

Projects

Ongoing projects

Untitled Camera Game: a photography simulator + visual novel in Unity with original songs written by me

PastelDices: a game jam project polished for publishing

Untitled Japanese App: a .Net MAUI app that parses and analyses Japanese sentences with the help of LLM (Gemini)

追忆 (Zhuīyì) - In Search of Lost Words

#Unity #C# #InteractiveEnvironment #HCI

A serious game that teaches to write Chinese characters

- Wacom graphic tablet and Nintendo Switch Joycon as input methods (keyboard and mouse are optional)
- Implemented the \$P recognizer for character recognition
- Implemented Bayesian Knowledge Tracing (BKT) for player modeling and level suggestion

SPY

#Unity #C# #SeriousGame #EntityComponentSystem #Tracing

Enhanced a game that teaches programming notions

- Added a level editor, a visual novel UI and a new map
- Used Competence-based Knowledge Space Theory (CbKST) to organize levels in a skill tree structure
- Used xAPI to trace user actions

Meta-Reinforcement Learning Project

#Python #TensorFlow #DeepLearning

Reproduction and analysis of the paper Prefrontal Cortex as a Meta-Reinforcement Learning System, which consists of the adaptation of a LSTM model with A2C in Tensorflow.

Multi-Agent Simulation of Ecotopia

Simulation of transport, housing, energy, and electricity in a society based on the book Ecotopia with GAMA by LIP6

Game Jams (cf. Game Portfolio)

#Unity #C# #UnrealEngine5 #C++ #Blueprint

More on my Dev Portfolio:

HCI Projet: Immersive Book, HCI Project: Gestures vs. Keyboard, Multi-Agent Project Dedale, Realtime Vision System for Drones